

PROBLEM SET

ICPC Asia EC

Online Contest II

2026 · 12 Problems · 5 Hours

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Problem A. All Closed

Time Limit: 1 seconds | Memory Limit: 512 MB
Standard Input / Standard Output

Sulfox the fennec fox gives you n sets $S_1, S_2, \dots, S_n$, each containing distinct integers between 0 and $2^m - 1$.

You may perform the following operation any number (possibly zero) of times: choose an integer x with $0 \leq x < 2^m$, and for every $i = 1, 2, \dots, n$, insert x into S_i if it is not already present.

A set S is called **XOR-closed** if, for every $x, y \in S$, we have $x \oplus y \in S$, where $\oplus$ denotes the bitwise exclusive-OR operation.

Find the minimum number of operations required to make all S_i XOR-closed, and construct any sequence of operations achieving this minimum.

Input

The first line contains two integers n ($1 \leq n \leq 2 \times 10^5$) and m ($1 \leq m \leq 20$).

The i -th of the next n lines contains an integer c_i ($c_i \geq 1$), followed by c_i distinct integers $a_{i,1}, a_{i,2}, \dots, a_{i,c_i}$ ($0 \leq a_{i,j} < 2^m$), representing the elements of S_i .

It is guaranteed that the sum of c_i does not exceed 10^6 .

Output

On the first line, print the minimum number of operations k .

On the second line, print k integers $x_1, x_2, \dots, x_k$, where x_i denotes the integer chosen in the i -th operation.

If there are multiple optimal solutions, output any of them.

Examples

Sample Input 1	Sample Output 1
3 3 1 1 3 0 1 7 6 1 2 3 4 5 7	3 0 7 6
Sample Input 2	Sample Output 2
1 4 4 11 5 14 0	0

Note

For the first example, after performing the shown operations, S_1 and S_2 become $\{0, 1, 6, 7\}$, while S_3 becomes $\{0, 1, 2, 3, 4, 5, 6, 7\}$. It can be proved that it is impossible to make all sets XOR-closed using less than 3 operations.

For the second example, the only set is already XOR-closed, so no operation is needed.

Problem B. Bread

Time Limit: 1 seconds | Memory Limit: 512 MB

Standard Input / Standard Output



Natsu is eating bread at Sensei's café when a piece of bread falls onto the floor and starts rolling.

The fallen bread can be regarded as a convex polygon in a two-dimensional plane. Its n vertices, listed in counterclockwise order, are $A_1, A_2, \dots, A_n$. The vertices are pairwise distinct, and no three vertices are collinear. The floor can be regarded as the horizontal line $y = 0$.

Initially, the bread touches the floor along edge (A_n, A_1) . Let the position of vertex A_n at this moment be the origin. The bread rolls in the positive direction of the x -axis; that is, $A_n = (0, 0)$ and $A_1 = (x_1, 0)$, where $x_1 > 0$. The bread then starts rolling. The rolling process can be abstracted as an infinite sequence of rotations performed in order. In the i -th rotation, the entire bread is rotated clockwise about the current position of vertex $A_{((i-1) \bmod n)+1}$ until $A_{(i \bmod n)+1}$ touches the floor for the first time, that is, until its y -coordinate becomes 0 for the first time.

As a romantic philosopher, Natsu considers the rolling of the bread to be a coincidental and romantic event. For the sake of this romance, after the k -th rotation, Natsu records the distance $L(k)$ rolled by the bread and the area $S(k)$ swept out by the bread. You need to calculate their ratio as k tends to infinity, namely:

$$\lim_{k \rightarrow +\infty} \frac{S(k)}{L(k)}$$

The distance $L(k)$ rolled by the bread after the k -th rotation is defined as the distance from the origin to the position of vertex $A_{(k \bmod n)+1}$ when the k -th rotation ends. The area $S(k)$ swept out by the bread after the k -th rotation is defined as the area of the union of all regions covered by the convex-polygon bread at all moments from the beginning of the first rotation to the end of the k -th rotation.

Input

The first line contains one positive integer T ($1 \leq T \leq 10^3$), denoting the number of test cases.

For each test case, the first line contains one integer n ($3 \leq n \leq 10^5$), denoting the number of vertices of the convex-polygon bread.

The next n lines each contain two integers x_i, y_i ($0 \leq |x_i|, y_i \leq 10^9$). The i -th line denotes the coordinates of vertex A_i when the bread has just fallen onto the floor.

It is guaranteed that y_1, x_n , and y_n are all 0, and that $x_1 > 0$.

It is guaranteed that vertices $A_1, A_2, \dots, A_n$ are pairwise distinct, no three vertices are collinear, and they form a convex polygon in counterclockwise order.

It is guaranteed that the sum of n over all test cases in one test file does not exceed 3×10^5 .

Output

For each test case, output one floating-point number on one line, denoting the answer. Your answer is considered correct if and only if its relative or absolute error compared with the standard answer does not exceed 10^{-9} .

Suppose that your answer is a and the standard answer is b . Your answer is considered correct if

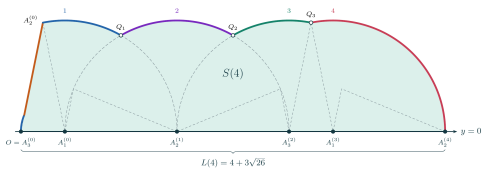
$$\frac{|a - b|}{\max\{b, 1\}} \leq 10^{-9}$$

Examples

Sample Input 1	Sample Output 1
2	4.908664519941197
3	1.384172075579563
2 0	
1 5	
0 0	
4	
1 0	
1 1	
0 1	
0 0	

Note

For the first test case, the process from the beginning of the first rotation to the end of the fourth rotation is shown below:



When $k = 4$, we have:

$$L(k) = 4 + 3\sqrt{26} \approx 19.297058541$$
$$S(k) = \frac{120}{13} + 13\sqrt{3} + \frac{91\pi}{6} + 43 \arctan \frac{1}{5} \approx 87.882927382$$

Problem C. Cut Tree

Time Limit: 10 seconds | Memory Limit: 512 MB
Standard Input / Standard Output

You are given a tree with n vertices. Vertex i has a weight z_i . The $n - 1$ edges are numbered from 1 to $n - 1$ in the order they are given in the input.

You have to answer q independent queries. A query is given by three integers l, r, f with $1 \leq l \leq f \leq r \leq n - 1$. For this query, build a graph on the n vertices that contains exactly the edges whose index lies in $[l, r]$, **except** that edge number f is deleted. Formally, the edge set is $\{i : l \leq i \leq r, i \neq f\}$.

This graph is a forest. The weight of a connected component is the sum of the weights of the vertices it contains (an isolated vertex forms a component of weight equal to its own weight). For each query output the maximum component weight.

Input

The first line contains one integer T ($1 \leq T \leq 10^5$), denoting the number of test cases. The description of the test cases follows.

The first line of each test case contains one integer n ($2 \leq n \leq 2 \times 10^5$).

Each of the next $n - 1$ lines contains two integers u_i and v_i ($1 \leq u_i, v_i \leq n, u_i \neq v_i$), denoting the two endpoints of edge i .

The next line contains n integers $z_1, z_2, \dots, z_n$ ($1 \leq z_i \leq 100$), denoting the vertex weights.

The next line contains one integer q ($1 \leq q \leq 2 \times 10^5$), denoting the number of queries.

Each of the next q lines contains three integers l, r, f ($1 \leq l \leq f \leq r \leq n - 1$) describing one query.

It is guaranteed that, within a single test file, the sum of n over all test cases does not exceed 2×10^5 , and the sum of q over all test cases does not exceed 2×10^5 .

Output

For each test case print q lines. The i -th of them must contain a single integer, denoting the answer to the i -th query of that test case.

Examples

Sample Input 1	Sample Output 1
2 5 1 2 2 3 3 4 4 5 10 20 30 40 50 3 1 4 2 2 4 4 1 1 1 2 1 2 7 5 1 1 1 1	120 90 50 7

Note

In the first test case the tree is a path $1 - 2 - 3 - 4 - 5$; edge i connects i and $i + 1$, and the weights are 10, 20, 30, 40, 50.

- Query (1, 4, 2): keep edges $\{1, 3, 4\}$. The component $\{1, 2\}$ has weight 30, and the component $\{3, 4, 5\}$ has weight 120; therefore, the maximum component weight is 120.
- Query (2, 4, 4): keep edges $\{2, 3\}$. The components $\{2, 3, 4\}$, $\{1\}$, and $\{5\}$ have weights 90, 10, and 50, respectively; therefore, the maximum component weight is 90.
- Query (1, 1, 1): no edge is kept, every vertex is isolated; the answer is the largest weight 50.

In the second test case the only edge is deleted by the query, so both vertices are isolated and the answer is $\max(7, 5) = 7$.

Problem D. Divide and Conquer

Time Limit: 2 seconds | Memory Limit: 512 MB

Standard Input / Standard Output

Segment tree is a data structure based on the divide-and-conquer algorithm, but this has nothing to do with this problem---it's just to make the title start with D.

For this problem, a segment tree is a perfect binary tree. A segment tree of depth n has exactly $2^{n+1} - 1$ nodes. Note that the root has depth 0, so a segment tree of depth n has $n + 1$ levels.

To make the structure of a segment tree easier to describe, we assign indices to its nodes. The root is assigned index 1. Every non-leaf node is assigned an index k satisfying $1 \leq k < 2^n$. It has exactly two children: its left child has index $2k$, and its right child has index $2k + 1$.

Each node of a segment tree also has a value. Let a_k denote the value of node k . For every non-leaf node k , its value a_k must be the maximum of the values of its two children; that is,

$$a_k = \max\{a_{2k}, a_{2k+1}\}.$$

The leaf nodes have indices $2^n, 2^n + 1, \dots, 2^{n+1} - 1$, and their values must form a permutation of $1, 2, \dots, 2^n$. In other words, the elements of the sequence $a_{2^n}, a_{2^n+1}, \dots, a_{2^{n+1}-1}$ must be pairwise distinct integers, and $1 \leq a_{2^n+i} \leq 2^n$ must hold for every $0 \leq i < 2^n$.

There are now q constraints. The i -th constraint is of the form (u_i, x_i) . Here, u_i is a node index and x_i is a specified value; the constraint requires the value a_{u_i} of node u_i to be equal to x_i . Calculate the number of segment trees satisfying all q constraints, or equivalently, the number of distinct valid sequences $a_1, a_2, \dots, a_{2^{n+1}-1}$. Output the answer modulo 998244353.

Input

The first line contains two integers n ($1 \leq n \leq 18$) and q ($0 \leq q \leq 2^{n+1} - 1$).

Each of the next q lines contains two integers u_i ($1 \leq u_i < 2^{n+1}$) and x_i ($1 \leq x_i \leq 2^n$), meaning that the value a_{u_i} of node u_i is required to be equal to x_i .

Output

Output one integer: the number of valid leaf permutations modulo 998244353.

Examples

Sample Input 1	Sample Output 1
2 1 2 2	4

Sample Input 2	Sample Output 2
2 2 1 4 2 4	12

Sample Input 3	Sample Output 3
2 2 2 3 4 4	0

Sample Input 4	Sample Output 4
4 2 1 2 1 3	0

Problem E. Exponent

Time Limit: 1 seconds | Memory Limit: 512 MB
Standard Input / Standard Output

The exponent of a group is the least common multiple of the orders of all elements in the group, but this has nothing to do with this problem. It's just to make the title start with E.

Little Ma gives you a positive integer n .

For each integer a with $1 \leq a \leq n$, define $\text{ord}_n(a)$ as the smallest positive integer x satisfying:

$$a^x \equiv 1 \pmod{n}$$

If no such positive integer x exists, define $\text{ord}_n(a) = 0$.

Compute:

$$\sum_{a=1}^n \text{ord}_n(a)$$

Input

The first line contains an integer T ($1 \leq T \leq 10^3$), the number of test cases.

Each of the next T lines contains one positive integer n ($1 \leq n \leq 10^9$).

Output

For each test case, output one integer in a single line: the required sum.

Examples

Sample Input 1	Sample Output 1
5 1 5 8 54 9653618	1 11 7 183 9856882167209

Sample Input 2	Sample Output 2
5 48 2 4 8 16	47 1 3 7 23

Problem F. Folding Game of Ohto Ai

Time Limit: 3 seconds | Memory Limit: 512 MB

Standard Input / Standard Output

Ohto Ai has an unrooted tree with N vertices. Let the current tree be $G = (V, E)$.

A **path** in a tree is a sequence of distinct vertices $v_1, v_2, \dots, v_k$ such that $\{v_i, v_{i+1}\} \in E$ for every $1 \leq i < k$.

A **diameter** of a tree is a path containing the maximum possible number of vertices.

While $|V| > 1$, Ohto Ai may perform the following **fold operation**: Choose any diameter $d_1, d_2, \dots, d_k$ of the current tree, listed in order from one endpoint to the other.

Define a function $f : V \rightarrow V$ by:

$$f(v) = \begin{cases} d_{\min\{i, k-i+1\}}, & \text{if } v = d_i \text{ for some } 1 \leq i \leq k \\ v, & \text{otherwise} \end{cases}$$

The tree after the fold is $G' = (V', E')$, where:

$$\begin{aligned} V' &= \{f(v) \mid v \in V\} \\ E' &= \{\{f(u), f(v)\} \mid \{u, v\} \in E, f(u) \neq f(v)\} \end{aligned}$$

In other words, vertices at symmetric positions on the chosen diameter are merged. Edges whose endpoints are merged into the same vertex disappear, and if multiple edges become edges between the same pair of vertices, only one copy is kept.

It can be shown that G' is still a tree.

Ohto Ai repeatedly performs fold operations until $|V| = 1$. Find the maximum possible number of operations.

Input

The first line contains an integer T ($1 \leq T \leq 3 \times 10^3$), denoting the number of test cases.

For each test case, the first line contains an integer N ($1 \leq N \leq 5 \times 10^4$), denoting the number of vertices in the tree.

The next $N - 1$ lines each contain two integers x, y ($1 \leq x, y \leq N, x \neq y$), denoting an edge of the tree.

It is guaranteed that the sum of N over all test cases does not exceed 5×10^4 .

Output

For each test case, output one line containing an integer: the maximum number of fold operations that can be performed on the tree.

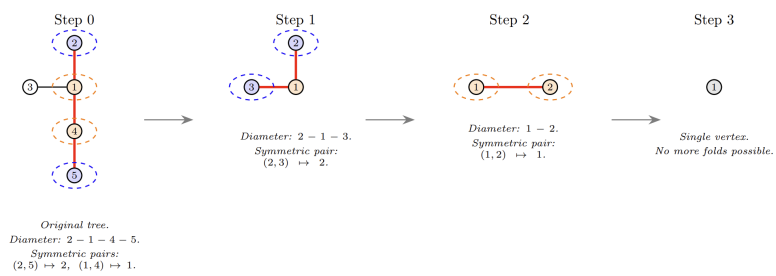
Examples

Sample Input 1	Sample Output 1
3 1 5 1 2 1 3 1 4 4 5 13 1 2 2 3 3 4 2 5 5 6 5 7 1 8 8 9 9 10 8 11 11 12 11 13	0 3 7

Note

For the first test case, the tree with 1 vertex has no fold operation that can be performed, so the answer is 0.

For the second test case, one optimal folding sequence is shown below:



For the third test case, it can be proven that at most 7 fold operations can be performed, so the answer is 7.

Problem G. Ghost of Tsushima

Time Limit: 4 seconds | Memory Limit: 512 MB

Standard Input / Standard Output



Background. Lately, Iochi Mari has had a small worry on her mind: as Antique Seraphim grows in popularity, her busy performance schedule leaves her little time to visit Schale. Meanwhile, Schale's Sensei misses her just as much. He therefore decides to rearrange her schedule, keeping the group's activities running smoothly while giving her some well-earned rest. Can you help poor Sensei count the schedules satisfying the conditions below?

Consider n vertices arranged in a cycle and n distinct edges $e_1, e_2, \dots, e_n$. For each $1 \leq i \leq n$, edge e_i connects vertex i to vertex $i + 1$, where vertex $n + 1$ is identified with vertex 1. Note that when $n = 1$, e_1 is a self-loop; when $n = 2$, e_1 and e_2 are two distinct parallel edges.

Intervals. For any $l, r \in \{1, 2, \dots, n\}$, the interval $[l, r]$ is defined as follows:

- If $l < r$, it covers vertices $l, l + 1, \dots, r$ and edges $e_l, e_{l+1}, \dots, e_{r-1}$.
- If $l > r$, it covers vertices $l, l + 1, \dots, n, 1, 2, \dots, r$ and edges $e_l, e_{l+1}, \dots, e_n, e_1, e_2, \dots, e_{r-1}$.
- If $l = r$, it covers only vertex l and no edges.

For example, when $n = 4$, the interval $[3, 2]$ covers vertices 3, 4, 1, 2 and edges e_3, e_4, e_1 .

We say interval A contains interval B if every vertex and every edge covered by B is also covered by A .

Valid sets. A set T of intervals is called valid if no interval in T contains another interval in T . The empty set is also valid.

Your task is to count valid sets under two different coverage limits. For a valid set T , define:

$$f(T) = \max_{\text{edge } e} |\{I \in T \mid I \text{ covers } e\}|,$$

$$g(T) = \max_{\text{vertex } v} |\{I \in T \mid I \text{ covers } v\}|.$$

For the empty set, define $f(\emptyset) = g(\emptyset) = 0$.

For each integer k from 1 to n , compute the following two counts separately:

1. The number of valid sets T satisfying $f(T) \leq k$.
2. The number of valid sets T satisfying $g(T) \leq k$.

Output all counts modulo 998244353.

Input

The first line contains a single positive integer T ($1 \leq T \leq 10^6$), representing the number of test cases.

Each test case consists of a single line containing one positive integer n ($1 \leq n \leq 10^6$).

It is guaranteed that the sum of n over all test cases in a single test file does not exceed 10^6 .

Output

For each test case, print two lines of n integers each.

- The k -th integer on the first line is the number of valid sets T satisfying $f(T) \leq k$.
- The k -th integer on the second line is the number of valid sets T satisfying $g(T) \leq k$.

All answers must be printed modulo 998244353.

Examples

Sample Input 1	Sample Output 1
3	2
1	2
2	7 7
4	6 7
	77 112 117 117
	46 99 116 117

Note

For $n = 1$, the only interval is $[1, 1]$, which covers vertex 1 and no edges. The two valid sets are $\emptyset$ and $\{[1, 1]\}$. Both satisfy $f(T) \leq 1$ and $g(T) \leq 1$, so both answers are 2.

For $n = 2$, the four intervals are $[1, 1]$, $[2, 2]$, $[1, 2]$, and $[2, 1]$. There are exactly 7 valid sets:

- the empty set;
- the four sets consisting of a single interval;
- $\{[1, 1], [2, 2]\}$ and $\{[1, 2], [2, 1]\}$.

Although $[1, 2]$ and $[2, 1]$ cover the same two vertices, they cover different edges: $[1, 2]$ covers e_1 , while $[2, 1]$ covers e_2 . Therefore, neither interval contains the other.

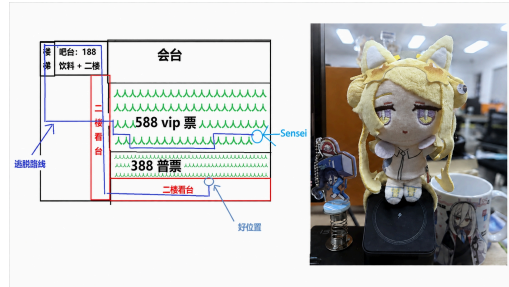
In every valid set, each edge is covered by at most one interval. Thus, the first output line is 7 7.

The set $\{[1, 2], [2, 1]\}$ covers each vertex twice. Every other valid set covers each vertex at most once. Thus, the second output line is 6 7.

Problem H. Hidden Track (Easy Version)

Time Limit: 2 seconds | Memory Limit: 512 MB

Standard Input / Standard Output



This is an interactive problem.

Background. On August 9, Sensei attends the *Hidden Track. Blue Archive 3rd Anniversary Concert* at FEISHENG LIVEHOUSE in Shanghai with Seia on his shoulders. Unfortunately, having arrived late, they are stuck in a remote corner until Seia spots a mysterious pay-to-win passage to the second floor. She knows the only route through the crowd, but the riddle-loving Seia refuses to reveal it directly. Instead, Sensei must find the hidden track by following her query rules within the given time and query limits. Can you help him reach the second floor before the passage disappears?

For each test case, the interactor hides a permutation $p = (p_1, p_2, \dots, p_n)$ of $\{0, 1, \dots, n-1\}$. When $n \geq 2$, it is guaranteed that $p_1 < p_n$. You may think of it as a hidden path, where p_i and p_{i+1} are connected by an edge for every $1 \leq i < n$. Your task is to find the hidden permutation using queries.

Let $k = \lceil \log_2 n \rceil$. In each query, you choose two integers m and v , where $0 \leq m < 2^k$, and either $v = -1$ or $0 \leq v < n$. The integer m determines a set of vertices:

$$B(m) = \{x \mid 0 \leq x < n, \text{popcount}(x \& m) \text{ is odd}\}$$

Here, $\&$ denotes the bitwise AND operation, and $\text{popcount}(x)$ denotes the number of 1-bits in the binary representation of x .

The query set S is constructed as follows.

First, let $S = B(m)$.

If $v = -1$, do nothing; otherwise, toggle whether v belongs to S : remove v if $v \in S$, and add v otherwise.

The interactor then counts the edges of the hidden path with **exactly one endpoint** in S . Let this number be:

$$c(S) = |\{i \mid 1 \leq i < n, |\{p_i, p_{i+1}\} \cap S| = 1\}|$$

The interactor returns $c(S) \bmod 3$.

For each test case, you may make at most nk queries, with $k = \lceil \log_2 n \rceil$ as defined above. You must reconstruct the hidden permutation p within this query limit.

The interactor is non-adaptive: the hidden permutation for each test case is fixed before the interaction begins and does not change throughout the interaction.

Input

At the beginning of the interaction, you are given an integer T ($1 \leq T \leq 10^4$), denoting the number of test cases.

For each test case, the interactor first gives you an integer n ($1 \leq n \leq 10^3$). After you output the answer for the current test case, the interaction proceeds to the next test case, if any.

It is guaranteed that the sum of n over all test cases in one test file does not exceed 10^4 .

Interaction

For each test case, you may perform the following two types of operations. The number of queries must not exceed nk , with $k = \lceil \log_2 n \rceil$ as defined above.

Query: Output a line of the form `? m v`, where $0 \leq m < 2^k$, and either $v = -1$ or $0 \leq v < n$. Here m, v must be integers. Then read an integer $r \in \{0, 1, 2\}$, whose value is $c(S) \bmod 3$.

Answer: Output a line of the form `! q1 q2 ... qn`, where $(q_1, q_2, \dots, q_n)$ must be exactly equal to the hidden permutation $(p_1, p_2, \dots, p_n)$.

Only operations of the form `? count` toward the query limit. If the interactor returns -1 at any point, your program must terminate immediately. Continuing the interaction after receiving -1 may result in an arbitrary verdict.

After printing a query or the answer, do not forget to output the end of the line and flush the output. Otherwise, you will get the verdict *Idleness Limit Exceeded*. To do this, use: `fflush(stdout)` or `cout.flush()` in C++; `System.out.flush()` in Java; `flush(output)` in Pascal; `stdout.flush()` in Python; see the documentation for other languages.

Examples

Sample Input 1	Sample Output 1
2	? 1 -1
	? 1 0
3	? 1 1
1	? 1 2
1	! 1 0 2
0	
2	! 0
1	

Note

For the first test case, the hidden permutation is $p = (1, 0, 2)$. The path has two edges: $\{0, 1\}$ and $\{0, 2\}$.

All four queries use $m = 1$, so each query starts with $B(1) = \{1\}$:

- `? 1 -1`: The set remains $S = \{1\}$. Only edge $\{0, 1\}$ has exactly one endpoint in S , so the interactor returns 1.
- `? 1 0`: Add vertex 0, giving $S = \{0, 1\}$. Edge $\{0, 1\}$ has both endpoints in S , while edge $\{0, 2\}$ has exactly one. Thus, the interactor returns 1.
- `? 1 1`: Remove vertex 1, giving $S = \emptyset$. Neither edge has an endpoint in S , so the interactor returns 0.
- `? 1 2`: Add vertex 2, giving $S = \{1, 2\}$. Both edges have exactly one endpoint in S , so the interactor returns 2.

The program then outputs `! 1 0 2`. Here, $k = 2$, so at most $nk = 6$ queries are allowed. The example uses only four queries.

For the second test case, $n = 1$, so the only possible permutation is (0) . The program outputs `! 0` without making any queries.

Problem I. Island

Time Limit: 2 seconds | Memory Limit: 512 MB

Standard Input / Standard Output

An ancient island suffered divine punishment, and the roads connecting its various regions were shattered.

The island can be regarded as an undirected graph containing n vertices, numbered $1, 2, \dots, n$.

Initially, the roads on the island form an ordered rooted tree rooted at vertex 1. The children of each vertex have a fixed left-to-right order, which is given in the input. In particular, the root has at least two children.

Starting from the root, perform a depth-first search, always recursively visiting the children of each vertex from left to right. Suppose there are k leaf vertices. In the order in which they are first visited, they are $p_1, p_2, \dots, p_k$. Here, a leaf vertex means a vertex of degree 1; since the root has at least two children, it cannot be a leaf vertex.

Besides the $n - 1$ roads in the tree, there are also $k - 1$ roads built between consecutive leaves. For every $1 \leq i < k$, the i -th road connects leaf vertex p_i and leaf vertex p_{i+1} . That is, all leaves are connected consecutively into a chain in the above DFS order, and there are $n + k - 2$ roads on the island in total.

When the divine punishment occurs, each road is independently destroyed with probability $\frac{1}{2}$, or retained with probability $\frac{1}{2}$, while all vertices still exist. The vertices and the remaining roads form several connected components. Let X be the number of connected components of the island. Please calculate the expected value of X . Here a connected component is defined as a maximal connected subgraph. Note that an isolated node also counts as a component.

It can be proven that the answer is a rational number. Suppose the answer in lowest terms is $\frac{a}{b}$. You only need to output a nonnegative integer x satisfying $0 \leq x < 998244353$ and $bx \equiv a \pmod{998244353}$, namely, the answer modulo 998244353.

Input

The first line contains a positive integer T ($1 \leq T \leq 10^3$), denoting the number of test cases.

For each test case, the first line contains an integer n ($3 \leq n \leq 10^5$), denoting the number of vertices on the island.

The next n lines describe the tree. The i -th line begins with an integer c_i ($0 \leq c_i < n$, $\sum c_i = n - 1$), denoting the number of children of vertex i . It is followed by c_i pairwise distinct integers $e_{i,1}, e_{i,2}, \dots, e_{i,c_i}$ ($2 \leq e_{i,j} \leq n$), giving the indices of all children of vertex i in order.

It is guaranteed that, within a single test file, the sum of n over all test cases does not exceed 3×10^5 .

Output

For each test case, output one integer on one line, denoting the answer modulo 998244353.

Examples

Sample Input 1	Sample Output 1
2 3 2 2 3 0 0 6 3 6 2 3 2 4 5 0 0 0 0	374341634 694091779

Note

For the first sample, if vertices encountered again while backtracking are listed as well, the depth-first traversal starting from the root is:

$$1 \rightarrow 2 \rightarrow 1 \rightarrow 3 \rightarrow 1$$

Thus, the leaves appear in the order 2, 3. Before the divine punishment, the island has the following three roads:

$$(1, 2), (1, 3), (2, 3)$$

The answer for the first sample is $\frac{13}{8}$.

For the second sample, the corresponding traversal order is:

$$1 \rightarrow 6 \rightarrow 1 \rightarrow 2 \rightarrow 4 \rightarrow 2 \rightarrow 5 \rightarrow 2 \rightarrow 1 \rightarrow 3 \rightarrow 1$$

The leaves therefore appear in the order 6, 4, 5, 3. Before the divine punishment, the island has the following eight roads:

$$(1, 6), (1, 2), (1, 3), (2, 4), (2, 5), (6, 4), (4, 5), (5, 3)$$

The answer for the second sample is $\frac{590}{256} = \frac{295}{128}$.

Problem J. Joker or Judger

Time Limit: 1 seconds | Memory Limit: 512 MB

Standard Input / Standard Output

As the chief judge of The 2026 ICPC Asia EC Regionals Online Contest (II), Mr. J has diligently finished preparing and reviewing all the problems. Today, the contest is finally about to begin!

The 2025 ICPC Asia East Continent Online Contest (II)

☆   [-828]

Since last year's contest was harshly criticized, Mr. J cares greatly about this year's reputation. Therefore, he has set himself a goal: this year's contest should not receive too many negative reviews. To quantify the negative impact of these reviews, Mr. J assigns a score to each type of negative review:

Type of negative review	Score
WrongProblem X	100
SameProblem X	30
UnreasonableProblemArrangement	10
UnreasonableLimitForProblem X	5
WeakTestsForProblem X	3
BadProblem X	1

If a negative review does not match any of the negative-review types in the table, its score is 0. Matching is case-sensitive, and the string must exactly match the corresponding form in the table. For a type containing X , X is a placeholder; when matching, it must be replaced by exactly one uppercase English letter from $\{A, B, \dots, L\}$.

If you need to copy the concrete string corresponding to a type of negative review, we recommend referring to the first test case in the sample.

Mr. J has an expected total negative-review score P in mind. If the total score of all negative reviews is greater than P , Mr. J is a Joker; otherwise, Mr. J remains a qualified Judger. You have now collected all n negative reviews. Determine whether Mr. J is a Joker or a Judger.

Input

The first line contains a positive integer T ($1 \leq T \leq 10^3$), denoting the number of test cases.

For each test case, the first line contains two integers n, P ($0 \leq n \leq 10^3, 0 \leq P \leq 10^5$), denoting the number of negative reviews and the expected total negative-review score, respectively.

The next n lines each contain a string S ($1 \leq |S| \leq 50$) consisting only of uppercase and lowercase English letters, denoting one negative review.

It is guaranteed that, within a single test file, the sum of n over all test cases does not exceed 10^3 .

Output

For each test case, if the total negative-review score is greater than P , output the string **Joker** on one line; otherwise, output the string **Judger** on one line.

Examples

Sample Input 1	Sample Output 1
5 6 0 WrongProblemA SameProblemA UnreasonableProblemArrangement UnreasonableLimitForProblemA WeakTestsForProblemA BadProblemA 6 105 WrongProblemA BadProblemC wrongProblemB WrongProblemN WrongProblemAA UnreasonableProblemArrangementExtra 7 49 SameProblemJ UnreasonableProblemArrangement UnreasonableLimitForProblemA WeakTestsForProblemC BadProblemH SameProblemj BadProblem 4 140 WrongProblemJ SameProblemA UnreasonableProblemArrangement BadProblemJ 2 1 BadProblemJ BadProblemJ	Joker Judger Judger Joker Joker

Note

For the first test case, the scores of the 6 negative reviews are 100, 30, 10, 5, 3, 1, respectively. Their total score is $149 > P$.

For the second test case, the scores of the 6 negative reviews are 100, 1, 0, 0, 0, 0, respectively. Their total score is $101 \leq P$.

For the third test case, the scores of the 7 negative reviews are 30, 10, 5, 3, 1, 0, 0, respectively. Their total score is $49 \leq P$.

For the fourth test case, the scores of the 4 negative reviews are 100, 30, 10, 1, respectively. Their total score is $141 > P$.

For the fifth test case, the scores of the 2 negative reviews are 1, 1, respectively. Their total score is $2 > P$. Note that duplicate negative reviews are also counted.

Problem K. K-MEX

Time Limit: 1 seconds | Memory Limit: 512 MB

Standard Input / Standard Output

Let a be an array of length n . Define $\text{mex}(a)$ as the smallest non-negative integer that does not appear in a . For example:

- $\text{mex}([4, 0, 2, 1]) = 3$
- $\text{mex}([-2, -4, 3]) = 0$

Define a k -transformation as follows: for each index i ($1 \leq i \leq n$), independently choose to either keep a_i unchanged or replace it with $k - a_i$. After all choices are made, we obtain a new array a' .

Define $k\text{-mex}(a)$ as the maximum possible value of $\text{mex}(a')$ among all arrays a' that can be obtained from a via a k -transformation.

You must answer q queries. Each query provides a non-negative integer k ; compute $k\text{-mex}(a)$ for each query.

To reduce the output size, you are only required to output the bitwise XOR (exclusive-OR) of the answers over all q queries.

Input

The first line contains a single integer T ($1 \leq T \leq 10^3$) --- the number of test cases.

For each test case:

- The first line contains a single positive integer n ($1 \leq n \leq 5 \times 10^3$), denoting the length of the array.
- The second line contains n integers $a_1, a_2, \dots, a_n$ ($0 \leq a_i \leq 10^9$), denoting the initial elements of array a .
- The next line contains a single positive integer q ($1 \leq q \leq 5 \times 10^5$), denoting the number of queries.
- Each of the next q lines contains a single non-negative integer k ($0 \leq k \leq 10^9$), denoting you must compute $k\text{-mex}(a)$ for this k .

It is guaranteed that $\sum n \leq 5 \times 10^3$ and $\sum q \leq 5 \times 10^5$ across all test cases in one test file.

Output

For each test case, output a single integer, denoting the bitwise XOR (exclusive-OR) of the answers to all queries in that test case.

Examples

Sample Input 1	Sample Output 1
2 3 0 1 3 3 1 2 5 9 0 0 1 1 2 3 3 4 5 7 4 5 6 7 8 9 10	3 8

Problem L. Loop

Time Limit: 1 seconds | Memory Limit: 512 MB

Standard Input / Standard Output

You have fallen into a Loop trap. To escape from the Loop trap, you need to solve the following puzzle:

You are given an integer sequence $a_0, a_1, \dots, a_{n-1}$ of length n .

Construct an $n \times n$ grid, with both row and column indices starting from 0. The 0-th column, from top to bottom, is $a_0, a_1, \dots, a_{n-1}$. Starting from the 1-st column, each column is obtained by cyclically shifting the previous column down by one cell.

Formally, if the number in row i and column j is $b_{i,j}$, then

$$b_{i,j} = \begin{cases} a_i & j = 0 \\ b_{(i+n-1) \bmod n, j-1} & j > 0 \end{cases}$$

You are at the upper-left corner $(0, 0)$. In each step, you may only move right or down, until you reach the lower-right corner $(n-1, n-1)$. Every grid cell visited by the path is counted, including the starting and ending cells.

Please find the minimum possible sum of the weights of the grid cells visited by the path.

Input

The first line contains one positive integer T ($1 \leq T \leq 10^3$), denoting the number of test cases.

For each test case, the first line contains one integer n ($1 \leq n \leq 10^5$), denoting the length of the integer sequence.

The second line contains n integers $a_0, a_1, \dots, a_{n-1}$ ($0 \leq a_i \leq 10^9$), denoting the integer sequence.

It is guaranteed that, within each test file, the sum of n over all test cases does not exceed 10^5 .

Output

For each test case, output one integer in one line, denoting the minimum path weight sum.

Examples

Sample Input 1	Sample Output 1
7 4 3 4 2 1 1 7 5 8 0 9 0 1 7 5 1 9 0 0 9 9 4 1 1000 1000 1 6 4 0 9 0 1 9 3 1000000000 1000000000 1000000000	13 7 20 30 7 24 5000000000

Note

For the first test case, the constructed grid is as follows:

3	1	2	4
4	3	1	2
2	4	3	1
1	2	4	3

A path with the minimum possible sum of visited cell weights is

$(0, 0) \rightarrow (0, 1) \rightarrow (0, 2) \rightarrow (1, 2) \rightarrow (1, 3) \rightarrow (2, 3) \rightarrow (3, 3).$

Its weight sum is $3 + 1 + 2 + 1 + 2 + 1 + 3 = 13$. It can be verified that no valid path has a weight sum less than 13.